

Fourier interpolation of the q -space TDSCHA Lanczos without high-order force-constant tensors

TDSCHA development notes — `qspace_interpolation` branch

July 2026

Abstract

We present a method to run the full time-dependent self-consistent harmonic approximation (TDSCHA) Lanczos response calculation on a fine uniform q -mesh that is *not* commensurate with the supercell of the stochastic ensemble, while the ensemble itself remains on the coarse commensurate mesh. The two-phonon sector of the response lives on the fine mesh, so a phonon at $\mathbf{q}$ decays into pairs of phonons at interpolated wavevectors, densely sampling the two-phonon continuum. Unlike the standard route (computing, centering and Fourier-interpolating the third-order force constants $\Phi^{(3)}$, as in `cellconstructor.Spectral`), the method never computes or stores a high-order tensor: interpolation is performed at the level of the *per-configuration Bloch fields* entering the stochastic estimator of the anharmonic vertices. The scheme costs $O(N_f)$ per Lanczos iteration, is exact at commensurate wavevectors by construction, and reduces to designable “stochastic centering” kernels that reproduce the minimal-image (zero-padded) centering of tensor interpolation. We derive the method, prove its equivalence with centered-tensor interpolation, validate it against direct fine-supercell calculations on an anharmonic chain model, and demonstrate agreement with the standard d_3 dynamic-bubble spectral functions of `cellconstructor.Spectral` at interpolated wavevectors.

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1 Background: the q -space TDSCHA Lanczos

The q -space TDSCHA Lanczos [1, 2] computes response functions by iterating the linearized TDSCHA superoperator $\mathcal{L}$ on a state vector

$$\psi = [R_\nu(\mathbf{q}_p) | a'_{\nu_1\nu_2}(\mathbf{q}_1, \mathbf{q}_2) | b'_{\nu_1\nu_2}(\mathbf{q}_1, \mathbf{q}_2)], \quad \mathbf{q}_1 + \mathbf{q}_2 = \mathbf{q}_p + \mathbf{G}, \quad (1)$$

with one $(n_b \times n_b)$ two-phonon block per pair of wavevectors summing to the perturbation wavevector $\mathbf{q}_p$ ($n_b = 3n_{\text{at}}$ bands per q -point). The harmonic part of $\mathcal{L}$ is diagonal and requires only the SSCHA frequencies and polarization vectors $\omega_\nu(\mathbf{q})$, $e_\nu(\mathbf{q})$. The anharmonic part is evaluated *stochastically*: for each configuration I of an importance-sampled SSCHA ensemble (weights ρ_I) one forms the mass-scaled Bloch fields

$$\begin{aligned} x_\nu(\mathbf{q}; I) &= \sum_a e_\nu^{a*}(\mathbf{q}) \sqrt{m_a} \tilde{u}_a(\mathbf{q}; I), & \tilde{u}_a(\mathbf{q}; I) &= \frac{1}{\sqrt{N_c}} \sum_{\mathbf{R}} e^{-i\mathbf{q}\cdot\mathbf{R}} u_a(\mathbf{R}; I), \\ y_\nu(\mathbf{q}; I) &= \sum_a e_\nu^{a*}(\mathbf{q}) \frac{1}{\sqrt{m_a}} \delta \tilde{f}_a(\mathbf{q}; I), & \delta f &= f - f_{\text{SSCHA}} - \langle f \rangle, \end{aligned} \quad (2)$$

where $\mathbf{R}$ runs over the N_c cells of the supercell and the phase convention follows `vector_r2q` (cell-origin phases). The estimator (the “fused kernel” of `tdscha_qspace.jl`) contracts these fields into weights and rank-one updates; every term is a product of three or four fields. Writing $r_1 = f_\Upsilon x$ with $f_\Upsilon(\mathbf{q}, \nu) = 2\omega/(1 + 2n)$ and $r_2 = y$, the three-field (D3) terms have the structure

$$\left\langle \overline{(f_\Upsilon x)}(\mathbf{q}_p) x(\mathbf{q}_1) y(\mathbf{q}_2) \right\rangle, \quad \langle \bar{y}(\mathbf{q}_p) x(\mathbf{q}_1) x(\mathbf{q}_2) \rangle, \quad \dots \quad (3)$$

and the four-field (D4) terms carry one more x . These combinations are *exact Gaussian integration-by-parts extractors* of the anharmonic vertices: for the Gaussian SSCHA density ρ_Υ ,

$$\langle u_a u_b \delta f_c \rangle = \Upsilon_{aa'}^{-1} \Upsilon_{bb'}^{-1} \langle \partial^3 V / \partial R_{a'} \partial R_{b'} \partial R_c \rangle, \quad (4)$$

with no perturbative truncation; the disconnected Wick contractions vanish because $\langle u \rangle = \langle \delta f \rangle = 0$ and $\langle u \otimes \delta f \rangle \propto \Upsilon^{-1}(\langle \partial^2 V \rangle - \Phi) = 0$ at SSCHA stationarity. **No third- or fourth-order force-constant tensor ever appears**: this is the property the interpolation exploits.

2 Off-grid Bloch fields and the effective interpolation kernel

2.1 Why the naive off-grid transform is not an interpolation

The configuration $u(\mathbf{R})$ is periodic with the supercell: it carries information only at the N_c commensurate wavevectors. Transforming its *infinite periodic repetition* at an incommensurate $\tilde{\mathbf{q}}$ gives exactly zero (a Dirac comb on the commensurate points), while the *finite* sum of Eq. (2) evaluated at $\tilde{\mathbf{q}}$ is implicitly a **rectangular window** over one period — a zero-padded estimate whose quality depends entirely on which periodic image of each correlation component receives which phase. In tensor language this is the familiar statement that Fourier interpolation of $\Phi^{(n)}$ is only meaningful after *centering*: reassigning each element to the minimal-distance periodic images and zero-padding beyond half the supercell, as done by `ForceTensor.Center()` before any interpolation. The construction below performs the analogue of centering *on the estimator*, without tensors.

2.2 The effective kernel of a windowed estimator

Let each field slot of a D3 term carry a real window $W(\mathbf{n})$ over the integer cell label $\mathbf{n}$ (zero outside one period; the plain estimator is $W \equiv 1$),

$$\tilde{x}_W(\tilde{\mathbf{q}}; I) \propto \sum_{\mathbf{n}} W(\mathbf{n}) e^{-i\tilde{\mathbf{q}} \cdot \mathbf{R}_{\mathbf{n}}} u(\mathbf{R}_{\mathbf{n}}; I). \quad (5)$$

Take a D3 product with slots (z, w, v) and exact momentum conservation $\tilde{\mathbf{q}}_1 + \tilde{\mathbf{q}}_2 = \tilde{\mathbf{q}}_p$ (guaranteed on a uniform fine mesh containing $\mathbf{q}_p$; note $e^{i\mathbf{G} \cdot \mathbf{R}} = 1$ makes the conservation exact, not approximate). Using translation invariance of the ensemble, its expectation is

$$\langle \tilde{z}(\tilde{\mathbf{q}}_p) \tilde{x}_w(\tilde{\mathbf{q}}_1) \tilde{y}_v(\tilde{\mathbf{q}}_2) \rangle = \frac{1}{N_c^{3/2}} \sum_{\delta_1 \delta_2} S_{z,w,v}(\delta_1, \delta_2) c_3(\delta_1, \delta_2) e^{-i\tilde{\mathbf{q}}_1 \cdot \delta_1 - i\tilde{\mathbf{q}}_2 \cdot \delta_2}, \quad (6)$$

where c_3 is the physical three-point correlation at *actual* (unwrapped) lattice differences, the sum runs over all periodic images, and the **effective interpolation kernel** is the (per-dimension separable) window triple correlation

$$S_{z,w,v}(\delta_1, \delta_2) = \sum_s z(s) w(s + \delta_1) v(s + \delta_2). \quad (7)$$

Two structural facts follow (verified numerically to machine precision):

1. *Partition of unity.* For the plain window, $\sum_{\text{images}} S(\delta + L\mathbf{n}) = L$ for every difference class per dimension of length L : at commensurate $\tilde{\mathbf{q}}$ all images share the same phase and the estimator **reduces exactly to the coarse one**.
2. *Tent bias off-grid.* The plain kernel assigns weight L – spread to each image: for $L = 4$, a nearest-neighbour correlation component leaks 25% of its weight onto an image three cells away, which enters with the wrong phase at incommensurate $\tilde{\mathbf{q}}$. The interpolation passes through the exact on-grid values but sags in between.

2.3 Equivalence with centered-tensor interpolation

Equation (6) states that the ensemble expectation of the off-grid estimator *is* the trigonometric (Fourier) interpolation of the real-space correlation function c_3 , with image weights S . If S equals the minimal-image indicator (ties split), the interpolated object is identical to what one would obtain by (i) forming the correlation tensor on the coarse lattice, (ii) centering it à la `ForceTensor.Center()`, and (iii) evaluating its Fourier sum at $\tilde{\mathbf{q}}$ – except that no tensor is ever formed, the contraction with the Lanczos vector being performed configuration-by-configuration at $O(N_I N_f n_b^2)$ cost. With the pre-filtering of Sec. 5 the interpolated correlation becomes (a contraction of) $\Phi^{(3)}$ itself, i.e. *the same object interpolated by `Spectral.py`*; the two routes then share the same locality assumption and the same accuracy class, which is the basis of the benchmark of Sec. 9.

3 Stochastic centering: designed multitaper windows

3.1 Exactly solvable case: pair kernels

For two-field (pair) kernels the minimal-image weights are reachable exactly with two rectangular windows: with $r_m(\delta) = \max(0, m - |\delta|)$ the autocorrelation of $\text{rect}(m)$, for even L

$$\frac{1}{2} [r_{L/2+1}(\delta) - r_{L/2-1}(\delta)] = \begin{cases} 1 & |\delta| \leq L/2 - 1 \\ 1/2 & |\delta| = L/2 \quad (\text{Wigner–Seitz tie}) \\ 0 & \text{beyond,} \end{cases} \quad (8)$$

including the correct tie splitting, and satisfying the partition of unity. This proves that sharp centering is representable by separable per-configuration windowing.

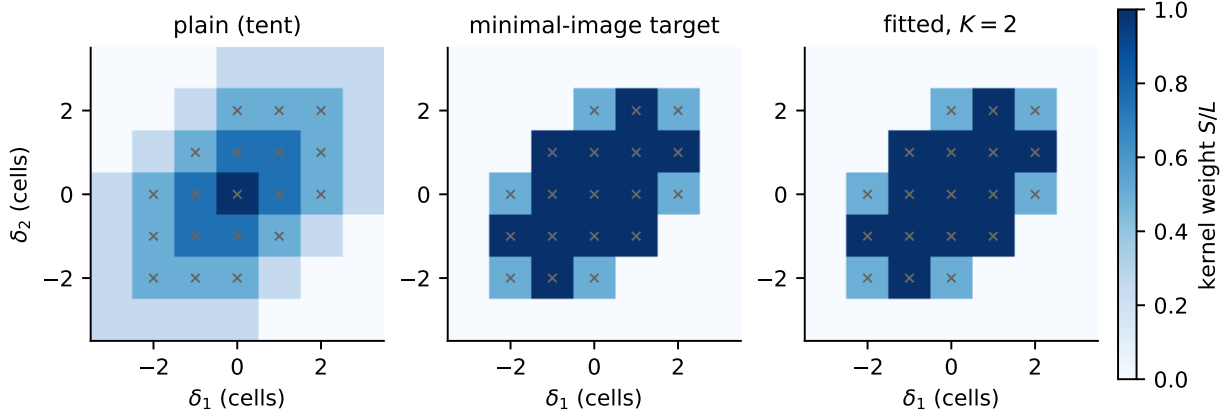


Figure 1: Effective interpolation kernels on the difference lattice (δ_1, δ_2) for a coarse dimension $L = 4$ (single hue: darker = larger weight; grey crosses mark the exact-image positions of the minimal-image target with their tie weights). Left: plain (tent) kernel of the naive off-grid transform. Middle: minimal-image target = zero-padded centering. Right: fitted $K = 2$ multitaper kernel — indistinguishable from the target (8×10^{-12} RMS).

3.2 Triple kernels: low-rank ALS design

The D3 kernel requires the two-argument target $M(\delta_1, \delta_2)$: the indicator of the minimal-spread image (spread = $\max(0, \delta_1, \delta_2) - \min(0, \delta_1, \delta_2)$), ties split — the separable analogue of the d_3 centering of **ForceTensor**. Same-window triples provably cannot improve on the tent kernel, but *slot-resolved* general windows have small canonical rank: per dimension we fit K passes (z_r, w_r, v_r) by alternating least squares so that

$$\sum_{r=1}^K \frac{1}{2} [S_{z_r, w_r, v_r} + S_{z_r, v_r, w_r}^T] \approx L M, \quad \text{subject to } \sum_{\text{images}} \sum_r S_r = L \quad (\text{hard partition constraint}). \quad (9)$$

The $w \leftrightarrow v$ symmetrization is realized at runtime by averaging the two pass orientations, which preserves the transpose symmetry of the two-phonon blocks and the Hermiticity of $\mathcal{L}$. The fitted kernels are numerically *exact* for the sizes relevant here ($L = 3, K = 3$: residual 2×10^{-16} ; $L = 4, K = 2$: 8×10^{-12}), and the partition constraint keeps the commensurate limit exact for any design (Fig. 1). In three dimensions windows and passes are per-dimension products (K^3 passes; each pass costs one kernel application). Origin averaging — realized by *rolling the configuration data* while window and phases stay fixed; a cyclic shift of the window itself would wrap its support with inconsistent phases — is an optional pure variance reduction. The D3 terms of $\mathcal{L}$ (the mutually adjoint pair $\{R^{(1)} \rightarrow d^2V \text{ map}, \alpha^{(1)} \rightarrow f \text{ map}\}$) run on the windowed passes; the D4 terms use a single plain-window pass (shorter range, smaller N_c/N_f weight).

4 Vertex renormalization $N_c \rightarrow N_f$

Mode-space vertices scale with the number of cells N as

$$d_3(\mathbf{q}_1\nu_1, \mathbf{q}_2\nu_2, \mathbf{q}_3\nu_3) = \frac{1}{\sqrt{N}} \delta_{\mathbf{q}_1+\mathbf{q}_2+\mathbf{q}_3, \mathbf{G}} \bar{\Phi}_3(\dots), \quad d_4 = \frac{1}{N} \delta_{\Sigma\mathbf{q}, \mathbf{G}} \bar{\Phi}_4, \quad (10)$$

with intensive $\bar{\Phi}_n$. The coarse-ensemble estimator delivers the $N = N_c$ normalization; the fine-mesh Lanczos is the operator of a hypothetical N_f -cell supercell and requires $N = N_f$. Hence all D3-type averages are rescaled by $s_3 = \sqrt{N_c/N_f}$ and all D4-type averages by $s_4 = N_c/N_f$ (arguments `scale3`, `scale4` of the Julia kernel). Consistency check on the one-loop bubble:

$\Sigma \sim \sum_{\text{pairs}} |d_3|^2 \chi$ has N_f pair terms of size $s_3^2 N_c^{-1} |\bar{\Phi}_3|^2 = N_f^{-1} |\bar{\Phi}_3|^2$ — intensive and identical to the direct fine calculation. The uniform per-order rescaling is exact up to disconnected Wick pieces $\propto \langle u \otimes \delta f \rangle$, which vanish at SSCHA stationarity; the ensemble should therefore be reweighted to the converged SSCHA dynamical matrix before interpolating. Empirically, *disabling* the rescaling makes the anharmonic renormalization overshoot by a factor $\simeq N_f/N_c$ (pinned by a regression test), confirming both its presence and magnitude.

5 Field pre-filtering: interpolating Φ -ranged objects

This step proved *essential*, not optional. By Eq. (4) the raw correlation carries a phonon-propagator dressing $f_\psi = (1 + 2n)/2\omega$ on *every displacement leg*:

$$\langle x(\mathbf{q}_1) x(\mathbf{q}_2) y(\mathbf{q}_3) \rangle = f_\psi(\mathbf{q}_1) f_\psi(\mathbf{q}_2) [\Phi^{(3)} \text{ contraction}]. \quad (11)$$

In real space the object being interpolated therefore decays with the *propagator* range (algebraic low- T tails from the acoustic $1/\omega \sim 1/|\mathbf{q}|$ non-analyticity), not with the short range of $\Phi^{(3)}$. On the chain benchmark this produced a systematic factor ≈ 0.5 deficit of the interpolated renormalization, shrinking only slowly with the coarse size. The cure exploits the structure of the kernel, which multiplies every displacement leg by $f_\Upsilon = 1/f_\psi$ anyway:

1. apply f_Υ on the coarse grid per configuration (exact there: it strips the dressing mode by mode), rebuild the filtered real-space configuration, and transform *that* at the fine wavevectors;
2. set the kernel tables $f_\Upsilon, f_\psi \rightarrow 1$ (**prefiltered** flag);
3. fold the exact fine-side $f_\psi(\tilde{\mathbf{q}}_1) \otimes f_\psi(\tilde{\mathbf{q}}_2)$ into the $\alpha^{(1)}$ blocks (the α -contracted legs also use the filtered fields).

On-grid this is an exact identity (the machine-precision back-compatibility test is unaffected); off-grid the interpolated correlations now decay with the range of the anharmonic force constants themselves — the same locality class as tensor interpolation. All polarization vectors and $\chi/f_\Upsilon/f_\psi$ factors are evaluated *analytically at the fine* $\tilde{\mathbf{q}}$ (from the centered, ASR-projected interpolation of the SSCHA dynamical matrix with the time-reversal gauge $e(-\tilde{\mathbf{q}}) = e^*(\tilde{\mathbf{q}})$), so no band-structure information is interpolated stochastically.

6 Acoustic sum rules

Centering breaks translational sum rules — this is why **ForceTensor** always follows **Center()** with **Apply_ASR()**. An uncorrected violation is amplified by the $1/\omega$ thermodynamic factors precisely at the near- Γ fine wavevectors the interpolation adds. In the stochastic scheme the ASR has a sharper form: it holds *per configuration* ($\sum_a \delta f_a = 0$ by Newton’s third law and the D2 ASR; zero mass-weighted centre of mass), with zero statistical noise. Windowing breaks exactly this (forces balance only over the whole supercell). Two measures are implemented:

- for (near-)uniform windows, the translation zero-modes are projected out of the fields with the correct mass metric: displacements lose the mass-weighted COM, force residuals lose the net force redistributed $\propto m_a$ (a plain mean subtraction would leak into optical modes for unequal masses). For the plain window this is exact and free.
- for the sign-oscillating designed windows the field-level projection is a rank-one modification of the effective window and would silently change the kernel away from the designed one (measured: a systematic $O(30\%)$ bias). It is therefore *not* applied there: the designed kernels reproduce minimal-image centering, whose residual ASR status is that of a centered

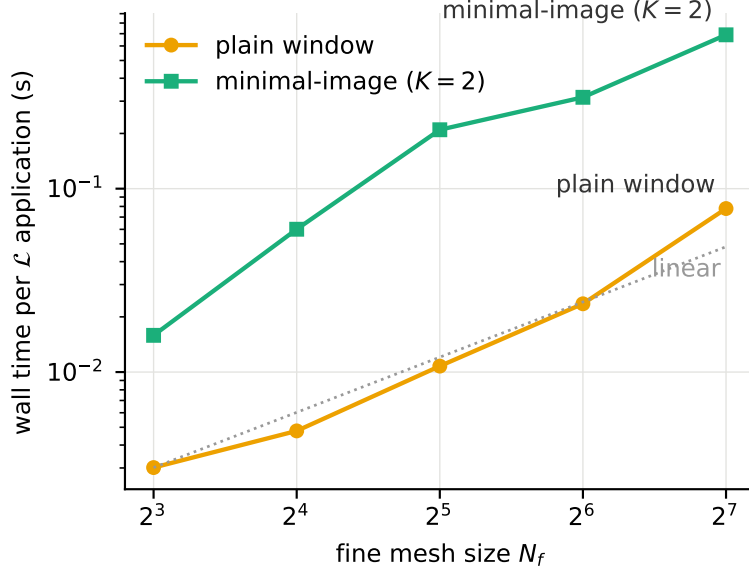


Figure 2: Wall time of one application of $\mathcal{L}$ versus the fine-mesh size N_f (chain toy, $N_I = 1000$ configurations, coarse $L_c = 4$): plain estimator and minimal-image windows ($K=2$, i.e. five kernel passes). Both follow the ideal linear trend (dotted).

(pre-Apply_AS_R) tensor. Folding the AS_R constraint into the kernel design itself (it is linear in the kernel) is left as future work; the interpolated dynamical matrix keeps its own exact D2 AS_R, and small-frequency guards protect the χ factors.

7 Implementation and cost

The implementation (branch `qspace_interpolation`) consists of `Modules/QSpaceInterpolation.py` (`QSpaceLanczosInterp(QSpaceLanczos)`: fine mesh with $O(1)$ hash lookups and $O(N_f)$ pair map, interpolated dynamical matrix, window design toolbox, field construction) and two Julia kernels in `tdscha_qspace.jl`: a slot-resolved scalar kernel and a batched BLAS-3 version (chunks of configurations at fixed symmetry; the symmetry rotation becomes one sparse $\times$ dense product per field, the $\alpha^{(1)}$ intermediates and the two-phonon accumulations become $(n_b \times n_b)(n_b \times B)$ and $(n_b \times B)(B \times n_b)$ GEMMs). The two kernels agree to 10^{-20} ; on the small toy ($n_b = 6$) they are equally fast (the sparse rotation dominates), the BLAS-3 structure being aimed at production band counts. Cost per Lanczos iteration: $O(K^3 N_I n_{\text{sym}} N_f n_b^2)$ — linear in the fine mesh, with K^3 the window-pass count ($= 1$ for the plain estimator). Measured wall times for one full application of $\mathcal{L}$ (Fig. 2) confirm near-linear scaling over $N_f = 8 \dots 128$.

8 Benchmarks

All benchmarks use an anharmonic diatomic chain defined by its topology (identical physics at every supercell length): harmonic springs K_1, K_2 per Cartesian component plus a per-bond cubic anharmonicity $g_3 s^3/3$ in the bond stretch, in the weak-coupling regime ($g_3 = 0.1$, renormalizations of $5 \dots 40 \text{ cm}^{-1}$ on $\sim 3000 \text{ cm}^{-1}$ modes). The harmonic bond force coincides with the SSCHA force, so the force residual is the exact anharmonic force. The triclinic cell has symmetry $P1$: no point-group operation can mask a convention error, and the mesh contains genuinely non-time-reversal-invariant wavevectors ($\mathbf{q} \neq -\mathbf{q}$).

Exactness checks. (i) With fine mesh = coarse mesh, the interpolated Lanczos reproduces the production `QSpaceLanczos` coefficients to 10^{-15} (with and without pre-filtering; through the

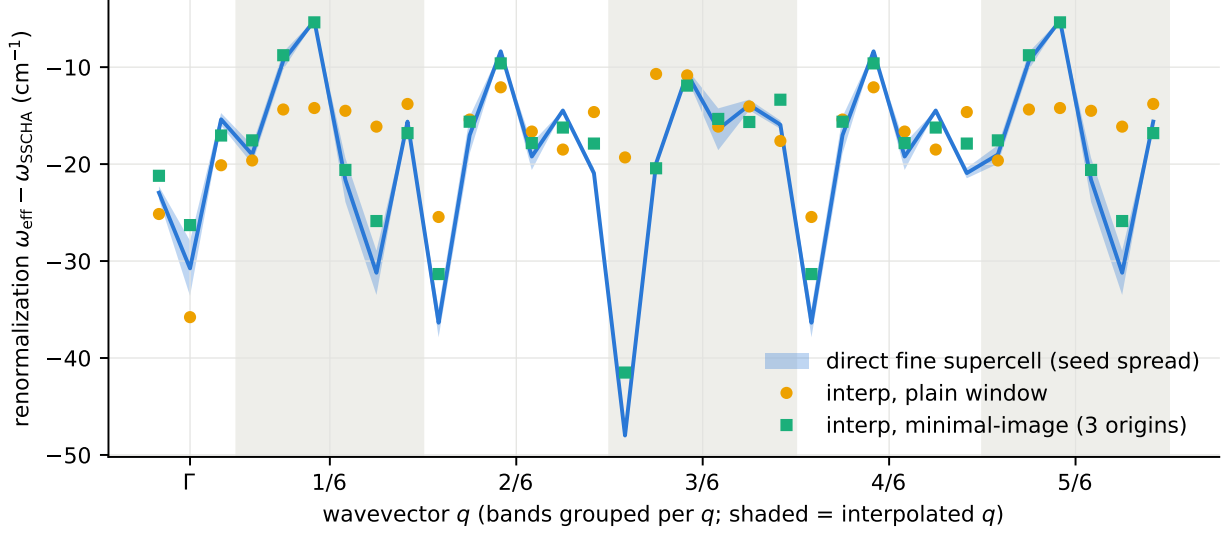


Figure 3: Mode-resolved anharmonic renormalization on the fine (1,1,6) mesh: direct fine-supercell reference (two independent seeds indicate the stochastic spread) versus the interpolated Lanczos from the coarse (1,1,3) ensemble with the plain window and with the designed minimal-image windows (3 origins). Vertical guides mark the interpolated (non-commensurate) wavevectors, including the non-TRI pair $q = \pm 1/6$.

batched kernel). (ii) The interpolated dynamical matrix reproduces the direct larger-supercell dispersion exactly for this strictly range-1 model. (iii) Rigid displacement shifts and constant force offsets leave the fields invariant (field-level ASR). (iv) The batched and scalar kernels agree to machine precision; the windowed path with trivial windows equals the plain path identically.

Physics: interpolation vs. direct fine supercell. Coarse (1,1,3) ensembles ($N_I = 4000$) interpolated to the (1,1,6) mesh are compared against *direct* (1,1,6)-supercell ensembles of the same model, mode by mode, on the anharmonic renormalization $\omega_{\text{eff}} - \omega_{\text{SSCHA}}$ (the static Lanczos response; absolute frequencies would hide the anharmonic signal). Table 1 and Fig. 3 summarize the aggregate error $\sum |\Delta_{\text{int}} - \Delta_{\text{dir}}| / \sum |\Delta_{\text{dir}}|$:

	aggregate error	comment
noise floor (two direct seeds)	0.103	statistical reference
plain window	0.299	tent-kernel bias
minimal-image windows, 1 origin	0.158	
minimal-image windows, 3 origins	0.122	at the noise floor
plain, <i>no</i> pre-filter	$\approx 2\times$ deficit	systematic (Sec. 5)
plain, <i>no</i> s_3, s_4 rescaling	$\approx 2\times$ overshoot	systematic (Sec. 4)

Table 1: Aggregate renormalization error of the interpolated Lanczos against direct fine-supercell calculations (chain toy, $L_c = 3 \rightarrow L_f = 6$, $N_I = 4000$, all 33 valid modes). The designed windows remove essentially all resolvable interpolation error; the last two rows document the failure modes that the pre-filter and the vertex rescaling correct.

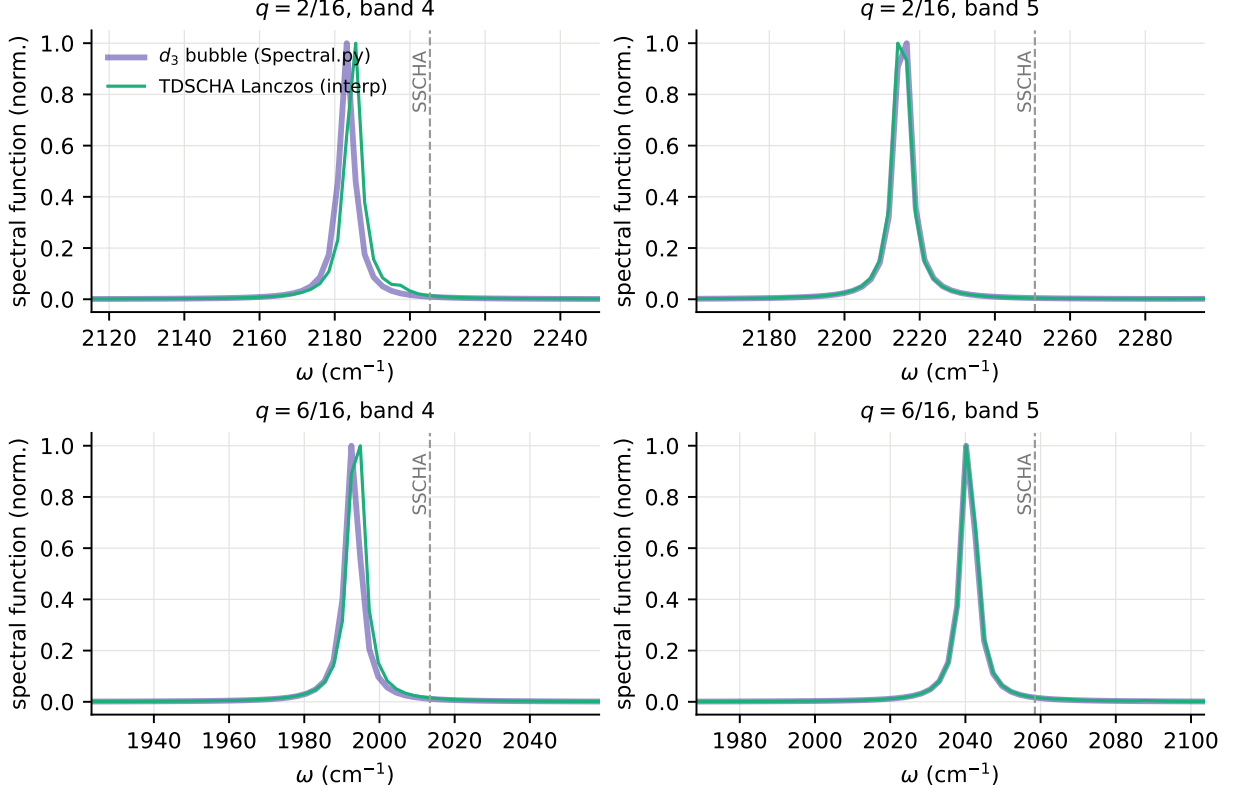


Figure 4: One-phonon spectral functions at interpolated wavevectors ($q = 2/16$ and $6/16$ along the chain, two optical bands): fine-mesh TDSCHA Lanczos interpolated from the coarse $(1,1,4)$ ensemble (minimal-image windows) versus the standard d_3 dynamic bubble of `cellconstructor.Spectral` with the exact centered $\Phi^{(3)}$ tensor on the same k -mesh and smearing. The SSCHA frequency is marked.

9 Example application: spectral functions at interpolated q and the standard d_3 bubble

As an end-to-end application we compute one-phonon spectral functions at wavevectors that do not exist on the coarse mesh and compare with the *standard* d_3 dynamic bubble of `cellconstructor.Spectral`, the third-order tensor route (here fed with the *exact analytic* $\Phi^{(3)}$ of the bond model, centered and ASR-projected, interpolated by `ForceTensor.Tensor3`), with the same k -integration mesh as the Lanczos fine mesh and the same smearing. The toy is purely cubic and weakly coupled, so the perturbative bubble and the (resummed) TDSCHA Lanczos should agree closely; the Lanczos runs with the D4 terms disabled to match the diagram content. Figure 4 shows the interpolated-Lanczos spectral functions on top of the bubble ones at interpolated wavevectors of the $(1,1,16)$ mesh from a coarse $(1,1,4)$ ensemble ($N_I = 8000$): peak positions and line shapes agree at the level of the energy-grid resolution (≈ 2.3 cm $^{-1}$), on anharmonic shifts of $15 \dots 40$ cm $^{-1}$:

q	band	ω_{SSCHA} (cm $^{-1}$)	peak (Lanczos interp) (cm $^{-1}$)	peak (d_3 bubble) (cm $^{-1}$)
2/16	4	2205.3	2185.6	2183.2
2/16	5	2250.6	2214.2	2216.6
6/16	4	2013.4	1994.9	1992.5
6/16	5	2058.5	2040.2	2040.2

10 Limitations and outlook

(i) Long-range (polar) parts of $\Phi^{(3)}$ are not separated analytically (the same limitation as the d_3 route of `Spectral.py`); LO-TO for the interpolated dynamical matrix is phase-2 work. (ii) The fine mesh adds no new microscopic information: noise at nearby fine wavevectors is correlated over the coarse-mesh spacing, and the genuine q -resolution of the *anharmonic* content is set by the coarse supercell — the physical gain is the dense two-phonon continuum and the interpolated harmonic dispersion. (iii) The ASR for the designed windows is inherited from minimal-image centering rather than enforced; folding it into the kernel design is future work. (iv) The distributed (MPI config-sliced) loader and the IR/Raman $\sqrt{N_f}$ prefactor for the interpolated class remain to be wired. (v) Off-mesh perturbation wavevectors (spectral maps along arbitrary paths with little-group symmetrization) are a natural extension: all required machinery (NUDFT at arbitrary $\tilde{\mathbf{q}}$) is already in place.

A Derivation of the effective interpolation kernel

We derive Eq. (6) in one dimension (the three-dimensional case is the per-dimension product). Let $u(s)$, $s \in \mathbb{Z}$, be the supercell-periodic configuration, $u(s + L) = u(s)$, and let the windowed field at wavevector $\tilde{q}$ (in units where the phase is $e^{-2\pi i \tilde{q}s}$) be

$$\tilde{x}_W(\tilde{q}) = \frac{1}{\sqrt{L}} \sum_s W(s) e^{-2\pi i \tilde{q}s} u(s), \quad (12)$$

with W supported on one period ($W = 1$ on $[0, L)$ for the plain estimator). Consider the three-field product with windows (z, w, v) on the slots and exact conservation $\tilde{q}_1 + \tilde{q}_2 = \tilde{q}_p$:

$$P = \overline{\tilde{z}(\tilde{q}_p)} \tilde{x}_w(\tilde{q}_1) \tilde{y}_v(\tilde{q}_2) = \frac{1}{L^{3/2}} \sum_{s_1 s_2 s_3} z(s_3) w(s_1) v(s_2) e^{2\pi i (\tilde{q}_p s_3 - \tilde{q}_1 s_1 - \tilde{q}_2 s_2)} \bar{u}(s_3) u(s_1) \delta f(s_2). \quad (13)$$

Taking the ensemble average and using translation invariance, $\langle \bar{u}(s_3) u(s_1) \delta f(s_2) \rangle = c_3(s_1 - s_3, s_2 - s_3)$ with c_3 evaluated at the *actual* integer differences (the physical correlation of the periodic system, which itself is the image sum $c_3(\delta_1, \delta_2) = \sum_{n_1 n_2} c_3^\infty(\delta_1 + n_1 L, \delta_2 + n_2 L)$ of the bulk correlation). Substituting $\delta_i = s_i - s_3$ and using $\tilde{q}_p - \tilde{q}_1 - \tilde{q}_2 = 0$ *exactly*, the s_3 dependence of the phase cancels:

$$\langle P \rangle = \frac{1}{L^{3/2}} \sum_{\delta_1 \delta_2} \underbrace{\left[\sum_{s_3} z(s_3) w(s_3 + \delta_1) v(s_3 + \delta_2) \right]}_{S_{z,w,v}(\delta_1, \delta_2)} c_3(\delta_1, \delta_2) e^{-2\pi i (\tilde{q}_1 \delta_1 + \tilde{q}_2 \delta_2)}. \quad (14)$$

Because the windows are zero-padded (support $\leq L$), the differences δ_i range over $(-L, L)$ and each *physical* image of c_3^∞ is weighted by S at its actual (unwrapped) difference and carries its actual phase — this is where the centering choice lives. Two consequences:

- **Commensurate limit.** If $\tilde{q}_i$ are commensurate, all images of a difference class $(\delta_1, \delta_2) \bmod L$ share the same phase, so only the class sums $\sum_{\mathbf{n}} S(\delta + L\mathbf{n})$ matter. If they equal L (partition of unity), Eq. (14) collapses to the exact coarse average — for *any* window design satisfying the constraint. The plain window satisfies it identically ($S = L - \text{spread}$, class sums $= L$).
- **Off-grid.** The kernel S/L is the image-assignment rule of the interpolation. The minimal-image target of Sec. 3.2 makes it the indicator of the most compact image (ties split), reproducing zero-padded tensor centering.

The same algebra applies verbatim to every D3 term (the y field can sit in any slot) and, with one more field, to the D4 terms, whose kernel is the window quadruple correlation. Note also that no origin appears in Eq. (14): translating the window (equivalently, rolling the data) leaves the expectation invariant, so origin averaging only reduces variance.

B Gaussian integration by parts and the pre-filter identity

For the Gaussian density ρ_Υ with covariance $\langle u_a u_b \rangle = \Upsilon_{ab}^{-1}$, Wick/Isserlis integration by parts gives, for any observable $F(u)$, $\langle u_a F \rangle = \Upsilon_{ab}^{-1} \langle \partial F / \partial u_b \rangle$. Applying it twice to the D3 average (with $\delta f = f - f_{\text{SSCHA}} - \langle f \rangle$, whose second derivative equals that of the full force $f = -\partial V / \partial R$):

$$\langle u_a u_b \delta f_c \rangle = \Upsilon_{ab}^{-1} \underbrace{\langle \delta f_c \rangle}_{=0} + \Upsilon_{aa'}^{-1} \Upsilon_{bb'}^{-1} \left\langle \frac{\partial^2 \delta f_c}{\partial u_{a'} \partial u_{b'}} \right\rangle = -\Upsilon_{aa'}^{-1} \Upsilon_{bb'}^{-1} \langle \partial_{a'b'c}^3 V \rangle. \quad (15)$$

In the SSCHA mode basis Υ^{-1} is diagonal with eigenvalues $f_\psi(\mathbf{q}, \nu) = (1 + 2n_{\mathbf{q}\nu}) / 2\omega_{\mathbf{q}\nu}$: every *displacement* leg of the correlation carries a propagator factor f_ψ , while force legs carry none. Since the Lanczos kernel multiplies every displacement leg by $f_\Upsilon = 1/f_\psi$ (the $r_1 = f_\Upsilon x$ fields), the combination it averages is the *bare* vertex contraction. The pre-filter of Sec. 5 simply moves the f_Υ multiplication *before* the off-grid transform: on the coarse grid $f_\Upsilon x$ is computed exactly (mode-diagonal), and its real-space transform is a configuration whose correlations with y are contractions of the short-ranged $\Phi^{(3)}$ (in the Cartesian basis, before any polarization projection — the projections are applied at the fine wavevectors analytically). Off-grid, interpolating the filtered fields therefore interpolates a Φ -ranged object; the exact fine-side factors $f_\psi(\tilde{\mathbf{q}})$ that the kernel formulas still require are supplied analytically (folded into the $\alpha^{(1)}$ blocks and set to unity in the kernel tables).

For the four-field averages the same expansion produces, besides the connected $\langle \partial^4 V \rangle$ term, disconnected pieces proportional to $\langle u \otimes \delta f \rangle = \Upsilon^{-1} (\langle \partial^2 V \rangle - \Phi)$, which vanish when Φ is the converged SSCHA matrix. This is the stationarity condition under which the uniform vertex rescaling of Sec. 4 is exact.

C Reproducibility

All numbers and figures in this report are produced by the scripts in `tests/test_interpolation/` (toy model and physics tests) and the benchmark/figure scripts referenced in the repository; the decisive physics comparisons are enforced as regression tests (`test_physics_interp.py`, `test_windows.py`).

References

- [1] L. Monacelli and F. Mauri, *Time-dependent self-consistent harmonic approximation*, Phys. Rev. B **103**, 104305 (2021).
- [2] TDSCHA q -space Lanczos implementation, `tdscha.QSpaceLanczos` (this repository).
- [3] R. Bianco et al., Phys. Rev. B **97**, 214101 (2018) (dynamic bubble conventions used by `cellconstructor.Spectral`).